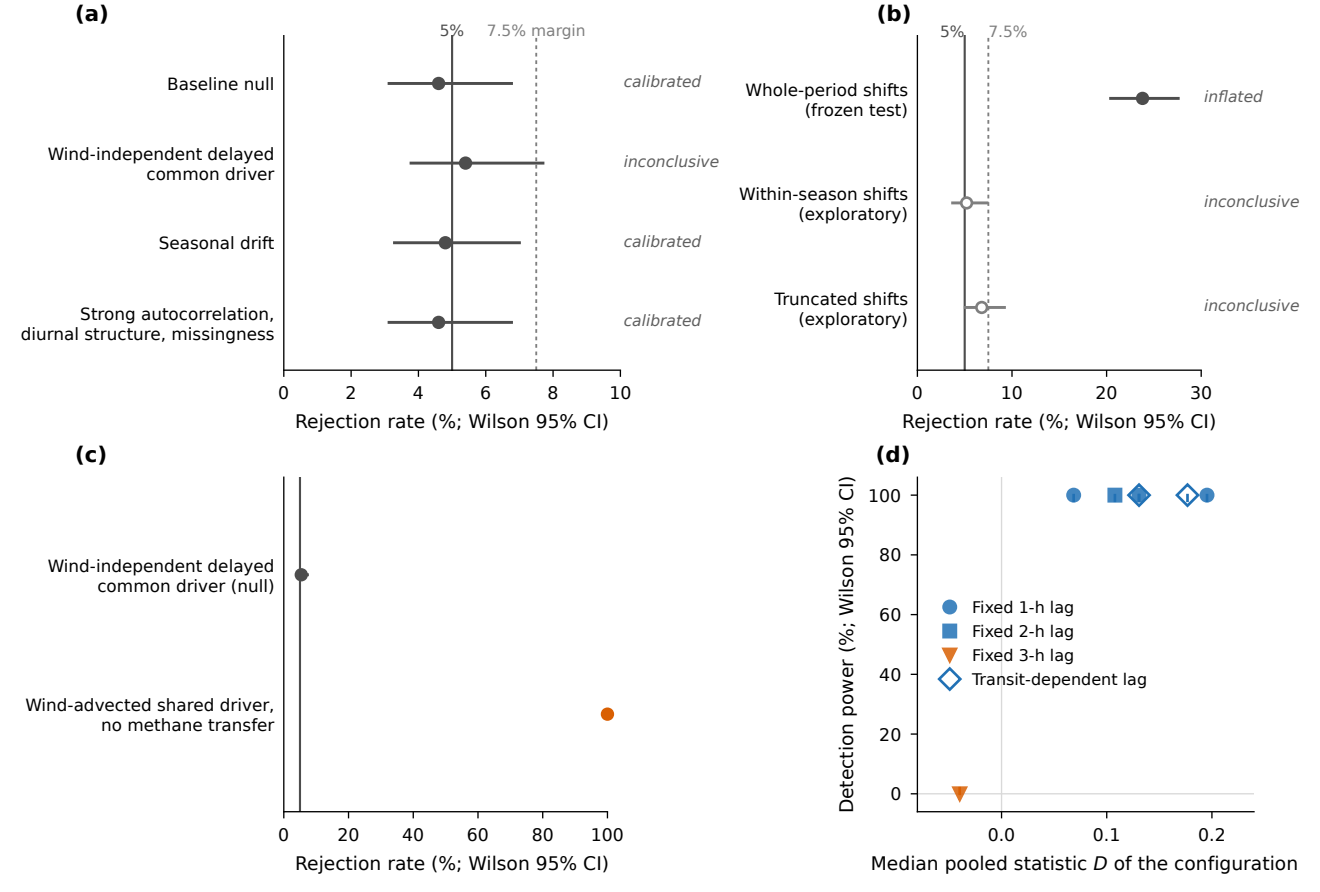
(a)
5%
7.5% margin
Baseline null
calibrated
Wind-independent delayed common driver
inconclusive
Seasonal drift
calibrated
Strong autocorrelation, diurnal structure, missingness
calibrated
0
2
4
6
8
10
Rejection rate (%; Wilson 95% CI)
(b)
5%
7.5%
Whole-period shifts (frozen test)
inflated
Within-season shifts (exploratory)
inconclusive
Truncated shifts (exploratory)
inconclusive
0
10
20
30
Rejection rate (%; Wilson 95% CI)
(c)
Wind-independent delayed common driver (null)
Wind-advected shared driver, no methane transfer
0
20
40
60
80
100
Rejection rate (%; Wilson 95% CI)
(d)
Detection power (%; Wilson 95% CI)
100
80
60
40
20
0
Fixed 1-h lag
Fixed 2-h lag
Fixed 3-h lag
Transit-dependent lag
0.0
0.1
0.2
Median pooled statistic D of the configuration